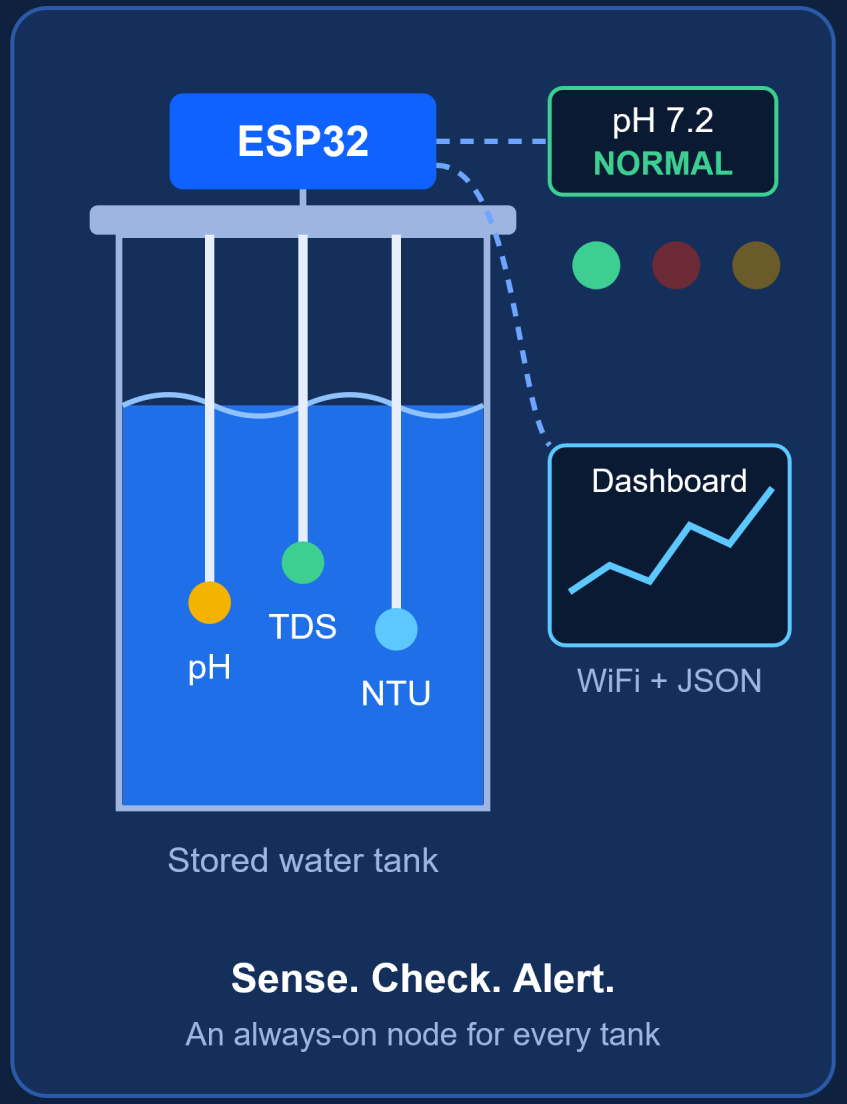
VIT CHENNAI | MULTIDISCIPLINARY ENGINEERING PROJECT | REVIEW 2

A Low-Cost Micro-Telemetry Architecture for Decentralized Water Quality Indexing Using I2C-Interfaced Visual Analytics

Early warning for stored water: continuous, at the source, between lab tests.

Abhinav Biju | Rohit Y | Doyel Dipak Chaudhari | Kartikey Rawat

Project guide: Dr. Aarthi B.



THE TEAM

One system, four complementary contributions

PROJECT GUIDE

Dr. Aarthi B.

VIT Chennai

First-year engineering team from multiple disciplines, combining hardware, embedded systems, software and data.

AB

Abhinav Biju 26BCE1252

Software and embedded: ESP32 code, sensor processing, thresholds, LCD/ I2C.

RY

Rohit Y 26BCE1255

Software and data/interface: display logic, integration, documentation.

DC

Doyel Dipak Chaudhari 26BEC1048

Hardware and electronics: sensor interfacing, circuits, assembly, testing.

KR

Kartikey Rawat 26BMH1036

Hardware and electronics: breadboard build, power, wiring, alert circuit.

THE PROBLEM

Water is tested when it arrives, not while it sits

Delivery
water arrives clean and tested

Lab test
checks one sample, days later

Everything in between
is guesswork

In the tank, quality can slip in three different ways.

1 Sediment and warmth

Sediment is stirred up and the tank warms.

Proxy signals: turbidity, temperature

2 Biofilm and fading disinfectant

Biofilm grows while disinfectant decays.

Proxy signals: ORP, TDS

3 Runoff, animals, leaks

Rain runoff, animals or pipe leaks get in.

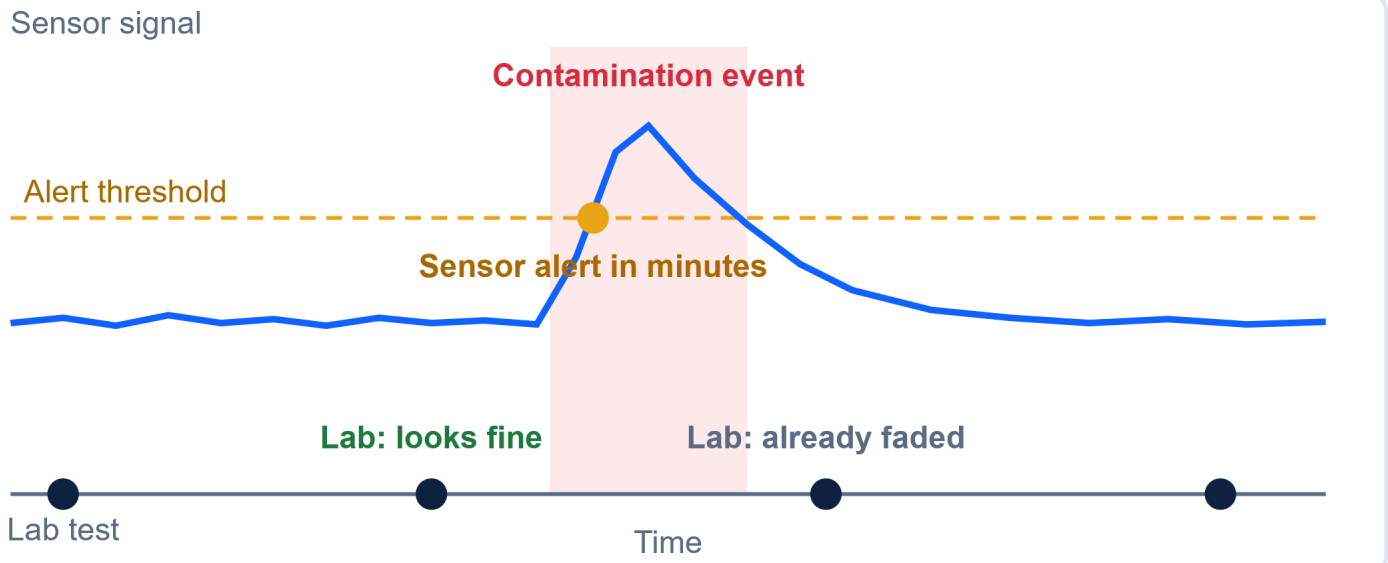
Proxy signals: turbidity, pH, TDS

Who feels it: hostels, schools, villages and apartment complexes in India that store water in overhead or underground tanks.

THE GAP

ILLUSTRATIVE

Detection delay: events start and end between tests



24–48 h

incubation for a bacterial test, before any result

Weekly, monthly or after complaints

how often samples are usually taken

A lab result is one moment. A sensor is the whole timeline. We watch what happens in between.

LITERATURE ANALYSIS

Who tests water today, and where it falls short

Who tests	What they do	Where it falls short
Utilities and municipal bodies	Test at plants and sample points on a schedule	Sparse and reactive
Public health labs, pollution boards	Test submitted samples	Days to weeks for results
Private NABL-accredited labs	Serve paying clients	Bacterial tests need 24–48 h
Rural schemes with field kits	Quick on-site checks	Patchy coverage, one-off samples

GAP WE ADDRESS

All are sparse, slow and reactive. Nothing watches the tank between tests, and lab-grade sensors still measure one sample, once.

PATENTS AND PAPERS

Next two slides: ten papers compared by topic and finding.

Low-cost monitoring: what exists and how reliable it is

Authors, year	Title	Topic	Research done and key finding
Lambrou, Anastasiou, Panayiotou, Polycarpou, 2014	A low-cost sensor network for real-time monitoring and contamination detection in drinking water distribution systems	**Contamination detection**	Low-cost sensor network aimed at real-time monitoring and detecting contamination in distribution systems (IEEE Sensors J.).
Pasika, Gandla, 2020	Smart water quality monitoring system with cost-effective using IoT	**Low-cost IoT monitoring**	Low-cost IoT monitoring system (Heliyon). Later work notes its alerts are basic threshold-based.
Jan, Min-Allah, Düstegör, 2021	IoT based smart water quality monitoring: recent techniques, trends and challenges for domestic applications	**Review: home use**	Reviews IoT techniques, trends and challenges for domestic water monitoring (Water, MDPI).
Bogdan, Paliuc, Crisan-Vida, Nimara, Barmayoun, 2023	Low-Cost Internet-of-Things Water-Quality Monitoring System for Rural Areas	**Rural sources, same sensor types**	Arduino UNO with DS18B20, pH, TDS and turbidity sensors and a phone app. Tested five rural sources in Romania; one exceeded 500 ppm TDS.
de Camargo, Spanhol, Slongo et al., 2023	Low-Cost Water Quality Sensors for IoT: A Systematic Review	**Sensor reliability**	Three main vendors; most sensors cost US$6.9 to US$169. Few papers compare low-cost sensors with reference devices.

Takeaway: low-cost sensing is well established, but few studies compare against reference devices, and most target sources, lakes or networks rather than stored tanks.

Reach and intelligence: LoRa, alerts and edge AI

Authors, year	Title	Topic	Research done and key finding
Jabbar, Ting, Hamidun et al., 2023	Development of LoRaWAN-based IoT system for water quality monitoring in rural areas	**Long-range links**	Solar-powered LoRa node with pH, TDS, turbidity and temperature. Tested at Gambang Lake; readings agreed with lab instruments.
Abrajano, Apanay, Botangen et al., 2024	IoT-Based Water Quality Monitoring System in Philippine Off-Grid Communities	**Community alerts**	Arduino system with pH, turbidity, temperature. Sends data to the cloud and SMS status messages to mobile numbers.
Mohammadi, Assaf, Assaad, Chang, 2024	An Intelligent Cloud-Based IoT-Enabled Multimodal Edge Sensing Device … Using Multisensor Data Fusion Technologies	**Multi-parameter fusion**	Fuses nine parameters, scores them with the NSF water quality index and shows results in a simple GUI.
Sharma, Mishra, Yadav, Gami, Madhan, 2026	An Intelligent, low-cost water quality monitoring system with on-device machine learning and cloud integration	**TinyML on ESP32**	Classifies normal, rainwater and chemical events at 99.28% on simulated profiles. In a 24 h test TDS drifted about 0.8 ppm per hour.
Bouziane, Aghoutane, Moumen et al., 2026	An intelligent and explainable IoT-Edge-Cloud architecture for real-time water quality monitoring	**Edge AI, explainability**	Computes and forecasts a water quality index at the edge; SHAP explains predictions. Validated on simulated sensor behavior (R^2 0.945).

Takeaway: edge AI and long-range links extend reach, but results come from simulated or short tests. Watching for change inside stored water between tests is still open.

OUR POSITIONING

A smoke detector, not the fire investigation lab

Always-on node
Crude but always present

→

Flags unusual change
“Go investigate”

→

Lab confirms
The step the system triggers

WHAT IT IS

- A continuous, cheap, at-the-source change detector
- Answers: has something unusual happened, and what should be done?

WHAT IT IS NOT

- Not a safety certifier: sensors measure proxies
- Cannot see bacteria, lead, arsenic, pesticides or fluoride

WE SAY

“No unusual change detected”

OR

“Unusual change detected, test recommended”

NEVER

“The water is safe”

OUR SOLUTION

From tank to decision

Tank sensors
pH, TDS, turbidity

→

ESP32
Averages, calibrates, alerts

→

JSON over WiFi
Raw + calibrated values

→

Backend
Stores history

→

Detectors + LLM
Find change, explain it

→

Alerts
Dashboard, WhatsApp, SMS

1 Sense smart

Many ADC samples averaged per reading to cut noise.

Raw values always kept

2 Calibrate yourself

pH probe calibrated with pH 4 and 7 buffers.

Recalibrate retroactively

3 Alert locally

Green or red LED and buzzer work even offline.

Safety rules on the device

Honest status: breadboard prototype with ESP32, three sensors, LCD, LEDs and buzzer is built. Backend, detectors and LLM interpreter are planned.

HARDWARE AND MODULES

Three sensing channels in one node

1 pH

Acidity or alkalinity: analog module and glass electrode.

2 TDS

Dissolved-substance response: analog module and probe.

3 Turbidity

Cloudiness from suspended particles: analog module.

PLANNED ADDITIONS

- DS18B20 temperature, ORP, I2C LCD backpack, voltage dividers

FORM FACTOR

- Lid-mounted, probes mid-depth, electronics outside the water

PROTOTYPE SIMULATION

SIMULATION

Circuit and pin connections





1 Inputs

pH, TDS and turbidity on GPIO 1, 2, 3 (ADC1).

Analog signals

2 Processing

ESP32-S2 averages many samples per channel.

Noise reduction

3 Outputs

I2C LCD, green and red LEDs, buzzer.

Local status

SOFTWARE

Firmware, data contract and backend

1 Firmware

C++ on Arduino framework (ESP32-S2). Libraries: Wire, I2C LCD [confirm].

Averaging, calibration, local rules

2 Backend (planned)

Python with Flask or http.server, SQLite, POST and GET /readings.

Minimal dependencies

3 Dashboard (planned)

One web page with Chart.js: pH and turbidity over time.

Then tank list, alerts

VERSIONED JSON MESSAGE

```
{
 "schema": 1,
 "device_id": "tank-hostel-01",
 "ts": "2026-10-09T10:15:00Z",
 "ph": 7.21, "ph_raw_mv": 2480,
 "turbidity_ntu": 3.4,
 "turbidity_raw_mv": 3100,
 "temp_c": 29.5, "tds_ppm": 312,
 "orp_mv": 410, "fw": "0.3.1"
}
```

Device and backend share only this format. Raw values are always stored.

ANALYSIS

Statistical detectors first, an LLM interpreter on top

Detector	What it catches
Rolling mean vs own baseline	Departure from normal
EWMA	Slow drift
Rate of change	Sudden spikes
Change-point detection	Shifts in behavior
Isolation Forest (later)	Unusual patterns

DERIVED FEATURES TO TEST

- Recovery time after inflow
- Settling rate in stagnation
- Disinfectant decay (ORP)
- Temperature-linked drift

LLM DIAGNOSIS EXAMPLE

“Tank 3: turbidity tripled since morning with no refill logged. Likely sediment disturbance or ingress. Recommend inspecting the lid and sending a sample for lab testing.”

FEASIBILITY

A low-budget build plan, checked against lab results

1 Trust sensors
Serial only: tap vs soil, vinegar vs soda

→

2 Tiny server
POST and GET / readings

→

3 Connect
ESP32 to server over WiFi

→

4 Chart page
pH and turbidity over time

→

5 Real tank
Bucket first, then a tank

KEEPING COST LOW

- Costs shared across the team
- Breadboard parts, raw code, minimal libraries
- SQLite and free tools, no paid cloud

RISKS AND MITIGATIONS

- Probe drift: regular calibration
- False alarms: tune against lab results
- Prototype on non-drinking water only

GROUND TRUTH LOOP

Lab samples are matched to the sensor timeline by collection time, labelling which patterns mean trouble.

FUTURE SCOPE

Ideas for the next phase

1 Mobile and web app

Tank status and alerts for operators.

Green, yellow, red

2 Smart water tanks

Lid nodes with solar or battery power.

GSM or LoRa for villages

3 Smart water bottle

A portable personal-check idea.

Concept only

4 Multi-node dashboard

Many tanks reporting to one view.

Wireless telemetry

5 Public status

Minimal QR code on the tank.

Operator-confirmed

6 Pipe bypass cell

A small sealed chamber for pipe monitoring.

Later version

VIT CHENNAI | MDP REVIEW 2

Stored water changes between tests.
We make that visible.

Detects

Unusual change in pH, TDS and turbidity, continuously.

Explains

Plain-language diagnosis and a recommended action.

Stays honest

Never says “safe”. The lab confirms.

Thank you | Questions and discussion

Abhinav Biju | Rohit Y | Doyel Dipak Chaudhari | Kartikey Rawat | Guide: Dr. Aarthi B.